Paper Title: DPy: Code Smells Detection Tool for Python
DOI: 10.5281/zenodo.14279535
Team Members: Bica Doru, Blezu Iosif, Bucur Mihnea

Tool Acquisition & Installation

I downloaded the DPy tool (as described in the paper) from its replication package. It came as a .zip archive containing an .exe file. Installation was straightforward—just extract the .zip and place the DPy.exe wherever desired. No additional configuration or dependencies were required.

Usage Experience

1. **Command Execution**:
   I opened a command prompt and ran:

   ```
   DPy.exe analyze -i path\to\my-python-project -o path\to\output
   ```
   This scanned all .py files in the specified folder.

2. **Output Generated**:
   DPy generated four JSON files:

   - {project_name}_class_module_metrics.json
   - {project_name}_design_smells.json
   - {project_name}_function_metrics.json
   - {project_name}_implementation_smells.json
   - logs file
     Each file contained detailed code metrics (e.g., LOC, cyclomatic complexity) or lists of detected smells (e.g., long identifiers, complex methods).

Benefits & Findings

- Easy Setup: No complicated environment needed; just run the .exe from a terminal.
- Comprehensive Metrics: The tool provided function-level metrics (LOC, complexity) and module-level metrics.
- Smell Detection: Highlighted both implementation (e.g., “Long identifier,” “Complex method”) and design smells (“Deficient encapsulation”).
- Actionable Insights: Quickly pinpointed a large method in my UI.py (high cyclomatic complexity), guiding a future refactoring plan.

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Screenshots

```
iosif@iosif MINGW64 ~/Desktop/VVSS-SEM
$ ./DPy.exe analyze -i ./iosif-blezu\ FP\ main\ A8/ ./out/
Usage: DPy.exe analyze [OPTIONS]
Try 'DPy.exe analyze --help' for help.
╭─ Error ──────────────────────────────────────────────────────────────────────╮
│ Missing option '--output' / '-o'.                                            │
╰──────────────────────────────────────────────────────────────────────────────╯

iosif@iosif MINGW64 ~/Desktop/VVSS-SEM
$ ./DPy.exe analyze -i ./iosif-blezu\ FP\ main\ A8/ -o ./out/
File not found: C:\Users\iosif\.designite\dpy.config
DPy version 1.3.0 Trial
Copyright (c) Designite 2025.
Current Configuration:
  Input Path: C:\Users\iosif\Desktop\VVSS-SEM\iosif-blezu FP main A8
  Output Directory: C:\Users\iosif\Desktop\VVSS-SEM\out
  Output Format: json
  Configuration File: C:\Users\iosif\AppData\Local\Temp\ONEFIL~1\default_config.
json
  Log Directory: C:\Users\iosif\Desktop\VVSS-SEM\out
File not found: C:\Users\iosif\.designite\dpy.config
Parsing ...
Calculating metrics ...
Detecting code smells ...
Detecting design smells ...
Analysis complete.
Preparing analysis summary ...
File not found: C:\Users\iosif\.designite\dpy.config
--Analysis summary--
        Total LOC analyzed: 378 Number of packages: 4
        Number of classes: 8    Number of modules: 11
        number of methods/functions: 79
-Total design smell instances detected-
        Multifaceted abstraction: 1     Deficient encapsulation: 3
        Broken modularization: 0        Insufficient modularization: 0
        Hub-like modularization: 0      Broken hierarchy: 0
        Deep hierarchy: 0       Wide hierarchy: 0
        Rebellious hierarchy: 0 Feature envy: 6
-Total implementation smell instances detected-
        Long method: 0  Long identifier: 4
        Long parameter list: 0  Complex method: 1
        Long statement: 0       Complex conditional: 0
        Empty catch block: 0    Magic number: 0
        Missing default: 0      Long message chain: 0
        Long Lambda Function: 0
Exporting results ...
Done.
```

dpy_log_20250326-223238	3/26/2025 10:32 PM	Text Document	3 KB
iosif-blezu FP main A8_class_module_metrics	3/26/2025 10:32 PM	JSON Source File	7 KB
iosif-blezu FP main A8_design_smells	3/26/2025 10:32 PM	JSON Source File	6 KB
iosif-blezu FP main A8_function_metrics	3/26/2025 10:32 PM	JSON Source File	19 KB
iosif-blezu FP main A8_implementation_smells	3/26/2025 10:32 PM	JSON Source File	3 KB

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```json
[
    {
        "Package": "iosif-blezu FP main A8.src.repository",
        "Module": "activity_repository",
        "Smell": "Deficient encapsulation",
        "Class/Module": "ActivityRepository",
        "Line no": "24",
        "File": "C:\\Users\\iosif\\Desktop\\VVSS-SEM\\iosif-blezu FP main A8\\src\\repository\\activity_repository.py",
        "Details": "Class 'ActivityRepository' accesses non-public code elements _Activity__date from update_activity"
    },
    {
        "Package": "iosif-blezu FP main A8.src.repository",
        "Module": "activity_repository",
        "Smell": "Deficient encapsulation",
        "Class/Module": "ActivityRepository",
        "Line no": "25",
        "File": "C:\\Users\\iosif\\Desktop\\VVSS-SEM\\iosif-blezu FP main A8\\src\\repository\\activity_repository.py",
        "Details": "Class 'ActivityRepository' accesses non-public code elements _Activity__time from update_activity"
    },
    {
        "Package": "iosif-blezu FP main A8.src.repository",
        "Module": "activity_repository",
        "Smell": "Deficient encapsulation",
        "Class/Module": "ActivityRepository",
        "Line no": "26",
        "File": "C:\\Users\\iosif\\Desktop\\VVSS-SEM\\iosif-blezu FP main A8\\src\\repository\\activity_repository.py",
        "Details": "Class 'ActivityRepository' accesses non-public code elements _Activity__description from update_activity"
    },
    {
        "Package": "iosif-blezu FP main A8.src.services",
        "Module": "person_service",
        "Smell": "Multifaceted abstraction",
        "Class/Module": "TestPersonService",
        "Line no": "48",
        "File": "C:\\Users\\iosif\\Desktop\\VVSS-SEM\\iosif-blezu FP main A8\\src\\services\\person_service.py",
        "Details": "Class 'TestPersonService' has a high LCOM value of 1, indicating it may have multiple responsibilities."
    },
```

```json
{
    "Package": "iosif-blezu FP main A8.src.ui",
    "Module": "UI",
    "Class": "UI",
    "Smell": "Long identifier",
    "Function/Method": "__list_all_activities_with_person",
    "Line no": "106",
    "File": "C:\\Users\\iosif\\Desktop\\VVSS-SEM\\iosif-blezu FP main A8\\src\\ui\\UI.py",
    "Details": "Identifier'__list_all_activities_with_person' exceeds the maximum length of 30 characters."
},
```


Conclusion:

DPy was simple to install and provided a rich set of metrics and smell detections, making it valuable for quickly identifying maintainability issues in Python code